

Enabling Virtual Reality Interactions in Confined Spaces by Re-Associating Finger Motions

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Figure 1: (a) We introduce the concept of re-associating body parts for VR interactions in confined spaces leveraging (b) small and less physically demanding finger motions to control a virtual arm. (c) We envision that the user can switch to finger mapping when using VR in confined spaces (e.g., toilet booth, vehicle) while still allowing for virtual arm interactions. Finger mapping is available for various interactions: (d) large swinging motions in slicing style games, (e) spatial selection and manipulation, and (f) locomotion in climbing based games.

ABSTRACT

As Virtual Reality (VR) headsets become mobile, people can interact in public places with applications often requiring large arm movements. However, using these open gestures is often uncomfortable and sometimes impossible in confined and public spaces (e.g., commuting in a vehicle). We introduce the concept of finger mapping, re-associating small-scale finger motions onto virtual arms in a larger VR space. Finger mapping supports various interactions (e.g., arms swinging movement, selection, manipulation, and locomotion) when the environment is constrained and does not allow large gestures. Finally, we discuss the opportunities and challenges of using finger mapping for VR interactions.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality

1 INTRODUCTION

Current Virtual Reality (VR) interactions are often based on an almost one-to-one mapping between the physical and virtual motion because

it allows users to easily understand the correspondence between translations and rotations on the input device. As VR Head-Mounted Displays (HMDs) are getting more mobile, users can immerse themselves wherever they wish. However, VR interaction requires enough space to be able to perform these large motions, and these new scenarios often happen inside of public and sometimes even confined spaces [1]. Using large and open gestures is usually undesirable in these environments since users could accidentally hit bystanders or do not even have enough space for interaction (e.g., seat inside a vehicle [2]).

In this paper, we present the concept of re-associating body parts [3, 5] to enable VR interactions in confined spaces. We introduce a finger mapping technique that maps small scale finger motions (Fig. 1ab) onto large scale virtual arm motions, enabling embodied VR arm and hand interaction in public and confined spaces (Fig. 1c). Finger mapping supports different VR interactions, such as large swinging motions (Fig. 1d), selecting and manipulating objects in space (Fig. 1e), and locomotion (Fig. 1f). Finally, we discuss the potential of using finger mapping as an alternative for input techniques, enabling virtual arm interactions in confined spaces and also working as a less fatigue option in VR.

2 INTERACTION DESIGN

Finger Mapping. The *Attach* mapping is based on homuncular flexibility that humans are flexible at learning a different relationship between tracked and rendered motion [5]. We see the virtual wrist position as a point within the user's maximum arm reach. The maximum arm reach is represented by a sphere that has an origin at the shoulder and a radius of the user's arm length (Fig. 2a). The virtual position of

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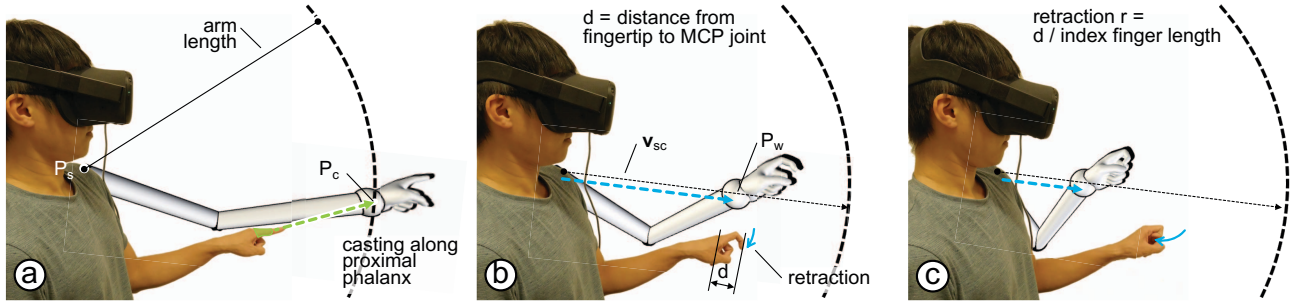


Figure 2: (a) *Attach* casts a ray along the proximal phalanx to a sphere, where the origin locates on shoulder and the radius is the user's arm length. (b)(c) When user bends his finger, we compute a retraction fraction with d to retract the virtual wrist towards shoulder.

wrist (P_c) is the intersection point of the sphere and a ray casting from the metacarpophalangeal (MCP) joint to the proximal interphalangeal (PIP) joint of the index finger. We choose the proximal phalanx since it is less intervened when bending the finger inwards. VR user either moves their proximal phalanx of the index finger or rotates the wrist to cast P_c on the surface of the maximum arm reach.

The virtual arm is fully stretched when using a pointing gesture (Fig. 2a). To reach positions inside the maximum arm reach sphere, we retract the virtual wrist along the direction between the casting point and shoulder by computing the retraction fraction r . r is the distance between the fingertip and MCP joint (d) divided by the length of index finger, which ranges from 0.15 to 1. The distance decreases as the user bends index finger inwards, and the virtual wrist retracts closer to the shoulder (Fig. 2bc). Therefore, the position of the virtual wrist (P_w) is computed as retraction amount multiplies by the arm length and the unit vector from the shoulder position to the casting point. The rotation of the virtual upper and lower arms are calculated by solving inverse kinematics using P_w as the target.

Selection. We design *ThumbButton* to support selection while using finger mapping. The user moves their thumb to touch the knuckle of middle finger, similar to pressing a button. We track the distance from the thumb fingertip to the middle finger knuckle (the PIP joint). The selection is triggered once detecting the thumb fingertip contacts the middle finger knuckle.

Misc. The current setup requires six-DoF tracking on the wrist, joints, and fingertips of the hand. We use Oculus Quest hand tracking to acquire the position and rotation data from both hands. The concept is also compatible with other VR systems providing similar hand tracking data. To increase some form of virtual body ownership and agency, we map the tracking information of physical fingers to virtual fingers. The virtual wrist orientation is the same as the physical wrist orientation. Although index fingers are occupied for positioning the virtual arm in space, the current implementation still allow finger interactions and simple manipulations at this new location.

3 DISCUSSION

We introduce the concept of finger mapping, re-associating finger motions to the virtual arm movement and enabling VR interaction in confined spaces. In our envision scenario, VR users can switch to finger mapping techniques when the environment is constrained and does not allow large motions. “A user started playing *BeatSaber* at their home but had to meet a friend. During the commute on the bus, the user sat in a confined public space and still wanted to continue the game. On the system level, the user changed the tracking boundaries to be stationary and selected to have ‘confined’ input. The VR HMD was now applying the finger mapping technique to map the user's hand motions to the virtual arms in the game.” Here, our approach is not yet another interaction technique that has to be implemented

to every application, but an abstract mapping that can be applied to any application working with hand tracking.

Finger mapping does not necessary have to come from being inside a confined space. Current VR interactions start to optimize for ergonomics to enable a longer and more comfortable experience [4]. Our approach can be an option when the user is tired and wants to have a lighter experience in terms of physical fatigue. This idea challenges the dominant notion of user needing to perform exactly the same type of actions as they do in real life inside VR (e.g., walking long distances). Considering a long gaming experience, users realize that it is not as much fun as it sounds to walk through all the virtual distances (e.g., *Skyrim*). Finger mapping aims to give users the feeling as if they would perform these interactions without actually performing them.

4 CONCLUSION

We present the concept of using finger mapping to enable arm interactions in confined spaces for VR, allowing users to control arm and hand interactions in VR by re-associating small finger motions. Finger mapping supports various functionalities, including large slicing motions, spatial selection and manipulation, and locomotion. Overall, we argue input techniques in VR should be similarly easy to adapt to the context as the tracking options are (e.g., room-scale and stationary tracking). The concept of finger mapping would work with any application when the environment is not allowed for full-body motions but is still available for spatial arm-based interaction in VR.

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